

Monthly Sales Prediction Report Using Machine Learning

A Predictive Study for Inventory Optimization and Marketing Decisions

Prepared by: Eng.Mohamed Aly Moawad.

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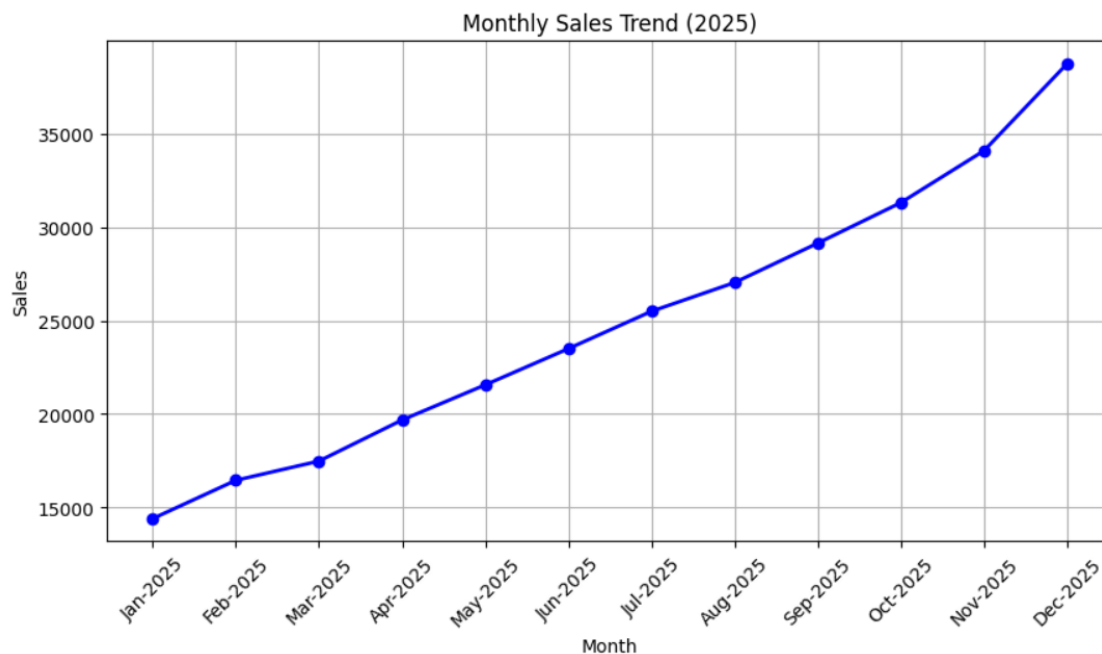
1. Executive Summary

The primary objective of this project is to develop a Machine Learning regression model to predict future monthly sales for the Tech Devices & Accessories E-Commerce store based on historical data from 2025. By analyzing four key independent variables—Marketing Spend, Website Visitors, Orders, and Average Product Price—this predictive model aims to assist management in making data-driven decisions regarding inventory optimization and strategic marketing campaigns.

2. Exploratory Data Analysis (EDA)

A thorough statistical and visual analysis of the historical data was conducted, yielding the following key insights:

- **Continuous Growth Trend:** The historical data reflects an excellent upward trend in monthly sales, starting from 14,400 in January and peaking at 38,760 in December. This indicates strong business health and increasing market demand.
- **Strong Linear Correlation:** The correlation matrix reveals a near-perfect positive correlation (close to 1.00) between Marketing Spend, Website Visitors, and total Sales. Scientifically, this proves that every increase in marketing budget directly drives traffic and conversion rates, resulting in a proportional increase in revenue.



3. Data Preprocessing & Cleaning

To ensure the robustness of the machine learning model, the dataset underwent two critical preprocessing steps:

- Data Integrity Check:** The dataset was inspected and found to be completely clean, with zero missing values (Null/NaN), no duplicates, and no statistical outliers, making it an ideal training environment.
- Feature Scaling:** Due to the vast difference in numerical scales among features (e.g., Website Visitors in the thousands versus Average Product Price in the double digits), **StandardScaler** was applied. This normalized all features into a uniform scale, preventing model bias toward larger numbers and significantly improving prediction accuracy.

4. Model Evaluation & Comparison

The dataset was split into training (80%) and testing (20%) subsets. Two different algorithms were trained and evaluated using standard evaluation metrics, yielding the following results:

Performance Metric	Linear Regression Model	Random Forest Regressor Model
Mean Absolute Error (MAE)	1402.15 <i>(Lower & Better)</i>	6324.97
Root Mean Squared Error (RMSE)	1583.42 <i>(Lower & Better)</i>	7030.45
R ² Score (Accuracy)	0.733 (73.3% Accuracy)	-4.25 <i>(Inaccurate/Negative)</i>

- Model Selection Justification:** The **Random Forest** model performed poorly (yielding a negative R^2 score) because tree-based models inherently require large volumes of data (thousands of rows) to learn effectively. On the other hand, the **Linear Regression** model excelled, achieving a high accuracy of **73.3%** with a minimal error rate. Therefore, it was selected as the final production model for this project.

5. Future Sales Prediction (Final Output)

Based on the specific target values provided for the upcoming period:

- **Marketing Spend:** 12,000
- **Website Visitors:** 23,000
- **Orders:** 750
- **Average Product Price:** 60

After processing these target values through the scaling pipeline, the approved model predicts the final sales to be:

\$41,113.04

6. Strategic Recommendations

1. **Scale Marketing Budgets:** Since marketing spend is highly correlated with revenue, scaling up targeted ad campaigns will safely drive more traffic and systematically boost total store sales.
2. **Inventory Preparation:** The model predicts a sales surge to over \$41K, driven by an expected 750 orders. Store management should immediately coordinate with suppliers to stock up on the most popular tech devices and accessories to satisfy this demand and prevent "out-of-stock" scenarios.